

Artificial Intelligence

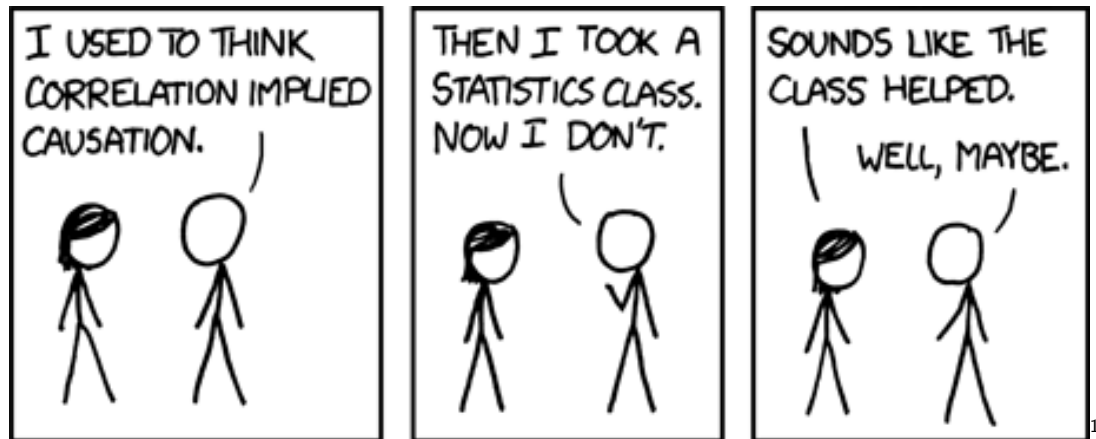
Causality

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Correlation Does Not Imply Causation



¹<https://xkcd.com/552/>

Nicholas Cage and Pool Deaths

The number of movies that star Nicolas Cage in a year is positively correlated with the number of pool deaths in that year.



Ice Cream and Shark Attacks



We know that ice cream doesn't cause shark attacks (right?), but perhaps there's something deeper.

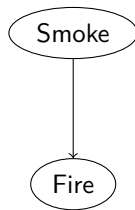
Where there's fire there's smoke.

Smoke and Fire are correlated – we tend to see them together. We can represent the full joint distribution over these variables with a Bayesian network.

Say we choose the ordering *Fire, Smoke*. Then we could use the Bayes net on the right and represent the full joint distribution as:

$$P(\textit{Smoke}, \textit{Fire}) = P(\textit{Smoke})P(\textit{Fire} \mid \textit{Smoke})$$

We learned earlier that using a Causal topological ordering gives us a more efficient representation.



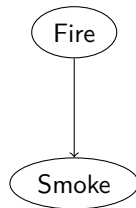
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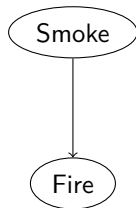
We learned earlier that while using a Causal topological ordering gives us a more efficient representation, any ordering gives us a valid representation of the full joint distribution.

- But does this network pass the smoke test?

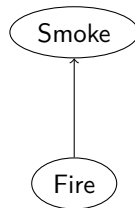


Causality Exists

We know that this is true:



and this is not true:



But probability theory and Bayes nets cannot represent this causal knowledge.

Probabilistic Causal Models

Causal networks are a restricted class of Bayesian networks which requires a causal topological ordering.

Example: Generating a Sample from Sprinkler Bayes Net

We have intervened with *do Sprinkler*, so the link from *Season* to *Sprinkler* is removed and $Sprinkler = true$, or $P(Sprinkler = true) = 1$.

